Extended essay: commentary for example essay			
DP subject: (Subject-focused essay)	Environmental Systems and Societies		
DP subjects: (Interdisciplinary essay)		Interdisciplinary framework:	
Research question:	How far is the enhanced infiltration of UVA radiation (315-400nm) due to ozone depletion less detrimental than UVB (280-315nm) for the germination and growth of Gerbera (Gerbera L.) and Chamomile (Matricaria chamomilla) flowering plants?		
Assessment details			
Criterion:	Mark awarded:	Commentary: (Incorporating wording from the assessment criteria)	
A: Framework for the essay [Maximum possible mark: 6]	**6**	The research question is relevant and focused. The methodology is appropriate for studying the research question and includes justifications for the selection of variables. For example, it explains the scientific reasons for selecting both plant species for this investigation. The essay is well-structured with some minor shortcomings, such as the use of inches instead of metric units.	
B: Knowledge and understanding [Maximum possible mark: 6]	**6**	The candidate uses a wide range of sources to support their assertions. The terminology is correctly used, demonstrating a solid understanding of the concepts within the systems and societies component. For example, the candidate discusses how the adverse impacts of climate change on plant growth vary across different global regions.	
C: Analysis and line of argument [Maximum possible mark: 6]	**5**	The candidate collects a substantial amount of relevant data and conducts a thorough analysis. The complexity of the research question, which involves measuring the effects of different radiations on the germination and growth of two different plants, makes it challenging to maintain a clear line of argument. Nevertheless, the candidate effectively observes general trends and analyzes individual differences.	
D: Discussion and evaluation [Maximum possible mark: 8]	**7**	The candidate discusses the significance of the results and attempts to explain or support them with data from other scientific research publications. However, to support some of the data, the candidate cites results from literature on topics beyond the scope of this research, such as metabolic changes. Additionally, some limitations, possible improvements, and suggestions for future research are presented.	

E: Reflection [Maximum possible mark: 4]	**4**	The candidate provides specific examples that illustrate clear learning experiences throughout the process. The reflection form also effectively demonstrates the student's growth and development.
Total marks awarded	28/30	
Holistic comment on the essay:	The candidate conducted a comprehensive laboratory investigation on the effects of different UV radiations on the germination and growth of two flowering plants. The research demonstrates a solid understanding of the relationship between environmental systems and societies. The research methods selected are very relevant and well-justified. The applied methodology enabled the collection of both qualitative and quantitative data, which were thoroughly explained and discussed. The essay concludes with a list of limitations, possible improvements, and suggestions for future research.	